NEUROLOGICAL LOCALISATION
SYMPTOM-BASED HISTORY & EXAMINATION

Bedside teaching reference · History → Examination → Anatomical structure

Curated by Anupam Singh MD, Professor, Department of Medicine, Santosh Medical College

Core rule: Describe the symptom precisely, identify the second neurological abnormality, then ask: *“What single anatomical lesion can explain all the findings?”* The tables below are designed as a pre-case framework before working through localisation cases.

Symptom Index — 14 Localisation Domains

1. Diplopia

2. Aphasia

3. Weakness — proximal vs distal

4. Gait abnormality

5. Numbness / sensory symptoms

6. Vertigo / dizziness

7. Dysarthria, dysphagia & dysphonia

8. Visual loss / field disturbance

9. Bladder, bowel & sexual dysfunction

10. Tremor & involuntary movements

11. Headache

12. Syncope / transient loss of consciousness

13. Seizure

14. Memory loss

01

Diplopia

Localise: ocular surface → extraocular muscle/NMJ → CN III/IV/VI → MLF/PPRF → cavernous sinus/orbital apex → brainstem

How to think: Start with monocular vs binocular. Then define horizontal/vertical separation, gaze dependence, ptosis, pupil change, facial numbness, visual loss and long-tract signs.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Diplopia persists when one eye is covered	Check each eye separately; inspect ocular surface/refraction	**Monocular diplopia — eye/refractive media**
Diplopia disappears when either eye is covered	Confirm ocular misalignment; test eye movements in all directions	**Binocular diplopia — EOM/CN III, IV, VI/ NMJ/brainstem**
Horizontal separation of images	Test abduction and adduction of each eye	**CN VI/lateral rectus, medial rectus/CN III, MLF or horizontal gaze pathway**
Vertical or oblique separation	Look for hypertropia; test elevation/ depression and head-tilt effect	**CN IV, CN III or central skew deviation**
Worse looking toward one particular side	H-pattern; identify the weak muscle in its field of action	**Individual EOM / corresponding ocular motor nerve**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Worse on reading or walking downstairs	Test depression of the adducted eye; Bielschowsky head-tilt pattern	**CN IV / superior oblique**
Ptosis with diplopia	Look at eye position and pupils	**CN III or NMJ; mild ptosis+miosis suggests sympathetic pathway**
Ptosis/diplopia fluctuates and worsens later in day	Sustained upgaze; repeated saccades; look for fatigability	**Neuromuscular junction — myasthenia**
Sudden headache with ptosis and diplopia	Look for complete III palsy and pupil enlargement	**Compressive CN III lesion; PCom aneurysm is an important cause**
Diabetes/HTN with painless diplopia and ptosis	III palsy with preserved pupillary response	**Microvascular CN III palsy**
Both eyes cannot look toward one side	Test conjugate horizontal gaze	**Ipsilateral PPRF or abducens nucleus — pons**
One eye fails to adduct; opposite eye has abducting nystagmus	Test horizontal gaze and convergence	**Ipsilateral MLF — internuclear ophthalmoplegia**
Diplopia + facial numbness	Test III/IV/VI and V1/V2 sensation/corneal reflex	**Cavernous sinus / parasellar region**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Diplopia + visual loss	Visual acuity, colour vision, RAPD plus ocular movements	**Orbital apex / optic nerve + ocular motor nerves**
Diplopia + contralateral weakness	Full cranial nerve and UMN examination	**Brainstem — midbrain or pons**
Diplopia + vertigo/ataxia	Nystagmus, skew, limb/gait coordination	**Brainstem/cerebellum / vestibular pathways**
Difficulty looking upward rather than isolated muscle weakness	Upgaze, convergence-retraction nystagmus, light-near dissociation	**Dorsal rostral midbrain — Parinaud region**

02

Aphasia

Localise the language network before naming the syndrome

How to think: First prove this is language dysfunction, not dysarthria, hearing loss, delirium or low consciousness. Test fluency, comprehension, repetition, naming, reading and writing.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Knows what they want to say but cannot get words out; short effortful answers	Nonfluent speech; relatively preserved comprehension; impaired naming/ repetition	**Dominant inferior frontal cortex — Broca/perisylvian frontal network**
Speaks fluently but family says speech makes no sense	Fluent paraphasic speech; poor comprehension and repetition	**Dominant posterior superior temporal cortex — Wernicke network**
Does the patient understand conversation?	One-step then multistep verbal commands	**Posterior temporal language comprehension network**
Difficulty finding names of familiar objects	Name pen/watch/body parts; describe object use	**Dominant temporal–temporoparietal naming network**
Fluent speech and comprehension good, but cannot repeat	Repeat phrases of increasing complexity	**Arcuate fasciculus/supramarginal region — conduction aphasia**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Nonfluent speech but repetition surprisingly preserved	Fluency ↓, comprehension relatively good, repetition preserved	**Dominant anterior watershed / medial frontal language network — transcortical motor**
Fluent speech, poor comprehension, but repetition preserved	Confirm preserved repetition despite poor comprehension	**Posterior watershed / temporoparietal association cortex — transcortical sensory**
Severe impairment of speech and comprehension	Nonfluent + poor comprehension + poor repetition	**Large dominant perisylvian/MCA territory — global aphasia**
Difficulty reading	Read aloud; obey a written command; test reading comprehension	**Dominant occipitotemporal/angular language pathway**
Cannot read but can still write	Ask spontaneous writing then ask patient to read it	**Left occipital cortex + splenium — alexia without agraphia**
Difficulty writing	Spontaneous sentence, dictated sentence and copying	**Dominant parietal/angular or broader dominant language network**
Calculation/right-left/finger identification difficulty	Acalculia, agraphia, finger agnosia, right-left disorientation	**Dominant angular gyrus — Gerstmann pattern**
Understands command but cannot demonstrate learned action	Pantomime brushing teeth, using a key or salute	**Dominant parietal–premotor praxis network**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Aphasia with face/arm weakness greater than leg	Right lower facial and arm UMN signs	**Dominant lateral hemisphere — usually left MCA cortex**

03 Weakness — proximal vs distal

Use function, distribution, tone, reflexes and sensation to move from muscle to nerve to cord to brain

How to think: Do not stop at “power 4/5”. Ask what daily task fails and whether weakness is proximal or distal, symmetric or asymmetric, fatigable, painful, spastic or associated with sensory loss.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Difficulty combing hair/reaching overhead	Shoulder abduction and external rotation; scapular movement	**Proximal upper limb — muscle or C5–C6**
Difficulty rising from chair without hands	Hip flexion/extension; sit-to-stand	**Proximal pelvic-girdle muscles / myopathy**
Difficulty climbing stairs or getting up from squat/floor	Hip and knee proximal power; Gowers-type manoeuvre	**Proximal lower limb / myopathy**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Difficulty buttoning, turning key, opening bottle	Finger abduction, opposition, grip, finger flexion/extension	**Distal upper limb — C8–T1/root/plexus/median-ulnar-radial**
Foot slaps or trips over toes	Heel walking; dorsiflexion and toe extension	**L4–L5/deep or common peroneal pathway**
Cannot stand on toes	Single-leg heel raise; plantarflexion; Achilles reflex	**S1–S2 / tibial nerve**
Symmetric proximal > distal weakness, no sensory loss	Proximal power, reflexes, muscle bulk	**Muscle / myopathy**
Proximal weakness + ptosis/ophthalmoplegia + fatigability	Sustained upgaze; repetitive proximal movements	**Neuromuscular junction — myasthenia**
Proximal weakness improves briefly after activity; dry mouth	Reduced reflexes with post-exercise facilitation	**Presynaptic NMJ — Lambert-Eaton**
Distal > proximal weakness + sensory symptoms	Distal sensory loss and reduced reflexes	**Peripheral neuropathy**
Weakness with neck/back radicular pain and dermatomal paresthesia	Myotomal power, reflex and dermatomal sensory testing	**Nerve root**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Several muscles from different nerves share one myotome	Map muscles by root rather than named nerve	**Radiculopathy**
Weakness fits one named nerve plus its cutaneous territory	Named-nerve motor and sensory testing	**Peripheral mononeuropathy**
Weakness crosses several named nerves with patchy sensory loss	Map root/nerve combinations	**Plexus**
Distal wasting/fasciculation without sensory loss	Fasciculations, atrophy, reflexes	**Anterior horn cell**
Wasting/fasciculation plus brisk reflexes/ Babinski	Look for UMN+LMN signs in multiple regions	**Motor neuron system**
Spastic weakness with brisk reflexes/ Babinski	Tone, reflexes, plantar responses	**Corticospinal tract — brain or spinal cord**
Face + arm + leg weakness on one side	Look for cortical signs and crossed cranial-nerve signs	**Contralateral hemisphere/internal capsule or brainstem**
Face/arm weakness greater than leg	Motor homunculus distribution; language/ neglect	**Lateral motor cortex — MCA territory**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Leg weakness greater than arm	Leg power, frontal signs, bladder symptoms	**Medial frontal motor cortex — ACA territory**
Dense proportional face-arm-leg weakness without cortical signs	No aphasia/neglect/field defect; pure UMN pattern	**Internal capsule**
LMN weakness/wasting in arms + UMN signs in legs	Segmental arm reflex loss + brisk legs/ Babinski	**Cervical spinal cord**
Bilateral UMN leg weakness, arms normal	Sensory level; brisk legs; Babinski	**Thoracic spinal cord**
Asymmetric painful flaccid legs + absent reflexes	Saddle sensation, root pattern, anal tone	**Cauda equina**
Dorsiflexion + eversion weak; inversion preserved	Test inversion and hip abduction	**Common peroneal nerve**
Dorsiflexion + inversion + hip abduction weak	L5 sensory pattern; ankle jerk often preserved	**L5 root**
Dorsiflexion and plantarflexion both weak	Hamstrings, ankle jerk and broad foot sensory loss	**Sciatic nerve**

04 Gait abnormality

Ask whether the gait problem is weakness, spasticity, ataxia, sensory loss, vestibular dysfunction, parkinsonism or frontal gait control

How to think: Describe the gait before naming it. Then test tone, reflexes, plantar responses, proprioception, Romberg, limb coordination, eye movements and distal power.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
One leg stiff and swings outward	Circumduction; flexed ipsilateral arm; brisk reflexes/Babinski	**Contralateral corticospinal tract — hemiparetic gait**
Both legs stiff; feet may cross	Spastic/scissoring gait; adductor tone ↑; brisk knees; Babinski	**Bilateral corticospinal tracts / spinal cord**
Feet slap; trips over toes	High-stepping gait; dorsiflexion weakness	**Distal LMN pathway / L5 / peroneal nerve**
Worse in darkness; watches feet	Stamping gait; positive Romberg; loss of vibration/joint position	**Posterior columns / large sensory fibres**
Broad-based, staggering, difficult turning	Wide base, irregular steps, impaired tandem walk	**Cerebellum**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Falls consistently toward one side	Limb dysmetria; nystagmus; vestibular testing	**Ipsilateral cerebellar/vestibular system**
Main problem is truncal instability, little limb incoordination	Truncal titubation; severe tandem impairment	**Cerebellar vermis**
Vertigo and veering to one side	Nystagmus; head impulse; gait veering/ falling	**Vestibular apparatus/CN VIII/brainstem**
Small steps, freezing, difficulty starting/ turning	Stooped posture; reduced arm swing; shuffling; en-bloc turning	**Basal ganglia — parkinsonism**
Feet seem 'stuck to floor'; cognition/ bladder symptoms	Magnetic gait; short steps; broad base; initiation difficulty	**Frontal-subcortical gait network**
Legs buckle; difficulty rising from chair	Proximal power; Gowers-type manoeuvre; waddling gait	**Pelvic-girdle muscle**
Pelvis drops on opposite side while standing on one leg	Trendelenburg sign; trunk lean toward weak side	**Gluteus medius/minimus / superior gluteal nerve / L5**
Radicular leg pain while walking	Root-specific weakness/reflex/sensory loss	**Radiculopathy**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Progressive stiff gait + urinary urgency	Spastic paraparesis + sensory level + Babinski	**Spinal cord myelopathy**
Unsteady because knees/feet cannot be felt	Positive Romberg; proprioceptive loss	**Dorsal columns / large-fibre neuropathy**
Unsteady even with eyes open	Dysmetria/dysdiadochokinesia/nystagmus	**Cerebellum**

05 Numbness / sensory symptoms

Map the sensory pattern before interpreting modality

How to think: The distribution often localises first: hemisensory, crossed face-body, sensory level, dermatome, named nerve, stocking-glove or saddle.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Face, arm and leg numb on the same side	Test all modalities over face and body	**Contralateral thalamus / sensory cortex**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Face one side + body opposite side	Pain-temperature over face and limbs	**Brainstem**
Numbness below a horizontal line on trunk	Map pinprick upward to define level	**Spinal cord**
Band-like chest/abdominal sensation	Dermatomal level; UMN leg signs	**Thoracic cord or root**
Shooting pain from back into limb	Dermatomal sensory loss + root-specific weakness/reflex loss	**Nerve root**
Numbness in one named nerve territory	Map cutaneous distribution	**Peripheral nerve**
Glove-and-stocking numbness	Distal pinprick/vibration loss + absent ankle jerks	**Length-dependent peripheral neuropathy**
Worse balance in darkness	Vibration, joint position, Romberg	**Posterior columns / large sensory fibres**
Burns/injuries without noticing temperature	Pain-temperature lost but vibration preserved	**Central cord / syrinx**
Saddle numbness	Perianal sensation, anal tone, sacral reflexes	**Conus / cauda equina**

06

Vertigo / dizziness

Separate peripheral vestibular disease from brainstem/cerebellar disease

How to think: Look for hearing symptoms, positional triggers, severe truncal ataxia and associated cranial-nerve or long-tract symptoms.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
True spinning sensation	Observe spontaneous nystagmus	**Vestibular system**
Sudden vertigo lasting hours-days with nausea/vomiting	Head impulse, nystagmus, skew	**Peripheral vestibular vs central brainstem/cerebellum**
Vertigo + hearing loss/tinnitus	Bedside hearing; otologic exam	**Labyrinth/CN VIII/AICA**
Seconds-long vertigo triggered by turning in bed	Dix-Hallpike	**Posterior semicircular canal — BPPV**
Unable to stand or walk despite little subjective vertigo	Truncal/gait ataxia	**Cerebellum**
Vertigo + diplopia/dysarthria/dysphagia	Cranial nerves, cerebellar and long-tract exam	**Brainstem / posterior circulation**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Vertigo + facial sensory loss + opposite body sensory loss	Crossed sensory examination	**Lateral medulla/pons**
Veers consistently to one side	Tandem gait; limb coordination	**Ipsilateral vestibular/cerebellar pathway**

07

Dysarthria, dysphagia & dysphonia

Differentiate language, articulation and voice

How to think: Aphasia = language. Dysarthria = articulation. Dysphonia = voice/larynx. Dysphagia may localise to corticobulbar, bulbar, NMJ or muscle.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Speech slurred but words/language appropriate	Articulation testing; repetition of lingual/ labial sounds	**Motor speech pathway — dysarthria**
Nasal speech + nasal regurgitation	Palatal elevation	**CN X / nucleus ambiguus**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Hoarse voice	Listen to phonation/cough; vocal cord examination if needed	**CN X / recurrent laryngeal**
Chokes mainly on liquids	Palate, gag, cough	**Oropharyngeal/bulbar pathway**
Food remains in mouth / hard to manipulate bolus	Tongue movements	**CN XII**
Weak cough	Cough strength	**CN X / bulbar muscles**
Tongue wasting/fasciculation	Inspect tongue at rest and protrusion	**LMN XII / bulbar motor neuron**
Brisk jaw jerk + slow stiff tongue without wasting	Jaw jerk, gag, limb UMN signs	**Bilateral corticobulbar — pseudobulbar**
Dysarthria + ipsilateral limb ataxia	Finger-nose, heel-shin	**Cerebellum/brainstem**
Dysarthria + clumsy hand, no aphasia	Fine hand movements and pyramidal signs	**Pons/internal capsule lacune**

Visual loss / field disturbance

Use monocular vs binocular and field geometry to map the optic pathway

How to think: One eye = retina/optic nerve. Bitemporal = chiasm. Homonymous = retrochiasmal.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Sudden painless monocular blindness	Visual acuity, RAPD, fundus	**Retina / optic nerve**
Pain on eye movement + colour desaturation	RAPD, red desaturation, central field	**Optic nerve**
Progressive peripheral visual loss in both eyes	Confrontation fields	**Optic chiasm**
Same side of visual world missing from both eyes	Homonymous field testing	**Retrochiasmal pathway**
Upper quadrant missing	Formal/confrontation fields	**Contralateral temporal Meyer loop**
Lower quadrant missing	Formal/confrontation fields	**Contralateral parietal optic radiation**
Hemianopia with otherwise normal neurologic exam	Congruity + macular sparing	**Occipital cortex**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Cannot recognise faces despite seeing clearly	Face recognition	**Fusiform cortex**
Cannot see but pupils remain reactive	Pupillary reflexes, fundus	**Bilateral occipital cortex**
Visual loss + ophthalmoplegia	RAPD + ocular movements	**Orbital apex**

09 Bladder, bowel & sexual dysfunction

These symptoms can sharply separate cord, conus and cauda

How to think: Always ask in bilateral leg weakness: difficulty passing urine? altered perianal/genital sensation? bowel or sexual dysfunction?

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Urgency/frequency with stiff legs	UMN legs + sensory level	**Spinal cord above sacral segments**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Sudden urinary retention + saddle numbness	Perianal sensation, anal tone, sacral reflexes	**Conus/cauda**
Very early bladder symptoms with relatively symmetric leg deficits	Sacral exam; mixed UMN/LMN signs	**Conus medullaris**
Severe radicular pain then later bladder retention	LMN legs; root sensory loss	**Cauda equina**
Loss of awareness of bladder filling	Sacral sensation/autonomic assessment	**Sacral cord/root pathways**
Erectile/sexual dysfunction + saddle sensory loss	S2–S4 sensory/autonomic assessment	**Conus / sacral roots**

10

Tremor & involuntary movements

First define the movement phenomenology; then localise the motor network

How to think: Rest tremor, intention tremor, ballistic movement, chorea, myoclonus and bradykinesia point to different networks.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Tremor when hand is resting	Observe hands at rest while distracted	**Basal ganglia / parkinsonian pathway**
Tremor appears as finger approaches target	Finger-nose testing	**Cerebellum**
Tremor during posture/action	Arms outstretched; writing/spiral	**Action tremor networks**
Violent flinging of one side	Observe proximal ballistic movements	**Contralateral subthalamic nucleus**
Irregular flowing purposeless movements	Observe face/limbs/tongue	**Striatum / chorea**
Brief shock-like jerks	Characterise distribution and stimulus sensitivity	**Myoclonus — cortex/subcortex/brainstem depending context**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Slowness rather than true weakness	Finger taps, pronation-supination, foot taps	**Basal ganglia / bradykinesia**
Abnormal movement disappears during sleep	Observation and history	**Supports hyperkinetic movement disorder**

11 Headache

Headache itself usually does not localise — associated neurological findings do

How to think: Define onset, speed to peak, position/Valsalva relation, fever, visual symptoms, focal deficits and cranial-nerve symptoms.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Sudden 'worst-ever' headache reaching maximum intensity within seconds/ minutes	Neck stiffness, consciousness, focal deficits	**Subarachnoid space / aneurysmal SAH**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Headache + focal weakness, aphasia or field deficit	Full cortical examination	**Cerebral hemisphere / vascular lesion**
Headache + vomiting + progressive visual blurring, worse in morning/coughing	Fundoscopy, CN VI, consciousness	**Raised intracranial pressure**
Headache worse on coughing/sneezing/straining	Papilledema, CN VI palsy, cerebellar signs	**Raised ICP / posterior fossa lesion**
Occipital headache + vomiting + gait unsteadiness	Gait, nystagmus, limb coordination	**Posterior fossa / cerebellum**
Headache + diplopia + multiple ocular motor palsies	III, IV, VI and V1/V2 examination	**Cavernous sinus / orbital apex**
Retro-orbital pain + visual loss	Acuity, colour vision, RAPD	**Optic nerve / orbit**
Headache + painful pupil-involving III palsy	Pupil and ocular movements	**PCom region / compressive CN III**
Headache + neck pain followed by Horner or posterior circulation symptoms	Horner signs; cranial nerves; cerebellar exam	**Carotid/vertebral artery dissection**
New unilateral headache + transient monocular visual loss in older adult	Temporal artery tenderness/pulse; visual acuity	**Temporal artery / ischemic optic pathway**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Headache + fever + neck stiffness	Meningism, consciousness, cranial nerves	**Meninges**
Headache + fever + focal deficit/seizure	Fundus and focal cortical examination	**Brain abscess/encephalitis/focal intracranial process**
Progressive headache + personality/ behavioural change	Frontal release signs, executive function	**Frontal lobe**
Headache + endocrine symptoms + visual-field loss	Bitemporal fields; ocular movements	**Sellar/suprasellar region**
Headache + papilledema + VI palsy without other focal signs	Fundoscopy and abduction	**Raised ICP**
Headache precipitated by posture/cough/ Valsalva + occipital symptoms	Lower cranial nerves and long-tract signs	**Foramen magnum / craniocervical junction**

12

Syncope / transient loss of consciousness

First decide whether this was true syncope, seizure or another transient neurologic event

How to think: True syncope reflects transient global cerebral hypoperfusion and usually does not localise to one lobe.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
LOC after prolonged standing, heat, pain or emotional trigger	Lying/standing BP, pulse	**Systemic autonomic/cardiovascular rather than focal CNS**
Prodrome of nausea, sweating, warmth, dimming vision	Orthostatic vitals; cardiovascular exam	**Vasovagal/autonomic pathway**
LOC immediately after standing	Orthostatic BP and pulse	**Autonomic nervous system / volume depletion**
Sudden LOC without warning during exertion	Cardiac examination; rhythm assessment	**Cardiac rather than neurological localisation**
Palpitations before LOC	Pulse/rhythm assessment	**Arrhythmic/cardiac source**
Brief LOC with rapid recovery and no confusion	Neurological exam usually normal	**Syncope rather than cortical seizure**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Brief jerking only after becoming unconscious	Look for absence of prolonged postictal focality	**Convulsive syncope can occur**
Tongue bite, prolonged confusion or stereotyped aura	Search for focal neurological signs	**Cortical seizure network**
LOC + diplopia/dysarthria/ataxia/bilateral weakness	Cranial nerves/cerebellar/long-tract exam	**Brainstem/posterior circulation**
LOC with persistent unilateral weakness afterward	Define fixed vs transient deficit	**Contralateral hemisphere; consider Todd paresis/stroke**
Recurrent syncope + impaired sweating/ erectile dysfunction/constipation	Orthostatic BP; autonomic exam	**Autonomic nervous system**

13

Seizure

The earliest ictal symptom is often the most localising

How to think: Ask 'what happened first?' before the seizure spread or generalised.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Jerking began in one hand and marched up arm toward face	Examine opposite motor cortex functions; look for Todd paresis	**Contralateral primary motor cortex**
Jerking began in face then spread to arm/leg	Postictal opposite weakness	**Contralateral motor cortex**
Forced head and eyes turned to one side at onset	Determine direction of version	**Usually frontal eye field opposite the forced gaze**
Tingling started in one hand and spread up arm	Cortical sensory testing	**Contralateral postcentral gyrus**
Flashing lights/coloured shapes	Visual fields	**Occipital cortex**
Complex visual scenes/familiar people before seizure	Visual recognition and temporal/occipitotemporal function	**Visual association cortex**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Rising epigastric sensation before impaired awareness	Memory/temporal examination	**Mesial temporal lobe**
Déjà vu, jamais vu or sudden intense fear	Memory/behaviour examination	**Mesial temporal lobe / amygdala**
Unpleasant smell or taste before seizure	Olfaction and temporal function	**Uncus / mesial temporal cortex**
Brief auditory hallucination or distorted sounds	Hearing/language exam	**Superior temporal cortex**
Speech arrest at onset	Language testing	**Dominant frontal/perisylvian cortex**
Cannot understand language during attack	Comprehension testing	**Dominant posterior temporal cortex**
Lip smacking, chewing or picking automatisms	Memory examination	**Temporal-limbic network**
Sudden tonic posturing of one limb	Look for focal UMN deficit afterward	**Contralateral frontal motor network**
Seizure followed by transient unilateral weakness	Document Todd paresis pattern and duration	**Opposite motor cortex involved during seizure**
Seizure followed by transient aphasia	Detailed language examination	**Dominant hemisphere**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Seizure followed by transient visual deficit	Visual-field testing	**Occipital/retrochiasmal cortex**
Behavioural arrest with staring	Automatisms, awareness and post-event confusion	**Temporal network if focal; generalized network if typical absence phenotype**
Bilateral tonic-clonic activity without obvious focal onset	Ask carefully for preceding focal aura	**May be focal-to-bilateral or generalized network**

14

Memory loss

Separate attention, registration, storage, retrieval and remote memory

How to think: Before calling it memory loss, exclude impaired attention, aphasia, delirium, depression and executive dysfunction.

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Repeatedly asks same question and forgets recent events	Delayed recall and new learning	**Hippocampus / medial temporal lobes**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Remembers childhood but cannot retain today's events	Immediate vs delayed recall	**Medial temporal memory consolidation system**
Cannot form new memories after an event	Learning of new words/objects	**Bilateral hippocampi**
Lost memories of events before illness	Establish temporal gradient of retrograde memory	**Distributed cortical memory stores / medial temporal access**
Forgetful with poor attention and fluctuating consciousness	Digit span; months backward	**Diffuse cortical/subcortical dysfunction**
'Forgetful' because patient cannot organise/plan/retrieve	Executive tests; verbal fluency	**Frontal-subcortical circuits**
Memory improves substantially with cues/ recognition choices	Free recall vs recognition	**Frontal retrieval dysfunction more than hippocampal storage**
Both recall and recognition poor despite adequate attention	Delayed recall and recognition	**Hippocampal storage/consolidation problem**
Memory loss + personality change/ disinhibition	Frontal and temporal examination	**Frontotemporal networks**
Memory loss + confabulation	Orientation, recall, executive control	**Diencephalic/frontal memory circuits**

Positive history findings / what to ask	Positive examination findings / what to elicit	Structure / localisation
Memory loss + gait ataxia + ophthalmoplegia	Eye movements, gait, confusion	**Mammillary bodies/medial thalamus — Wernicke network**
Sudden severe anterograde amnesia with preserved identity	Repetitive questioning; attention relatively preserved	**Bilateral hippocampal memory network / transient global amnesia phenotype**
Memory loss + vertical gaze palsy/ hypersomnolence	Consciousness and vertical eye movements	**Paramedian thalami**
Verbal memory poor; visual memory relatively preserved	Compare verbal learning with visual memory	**Dominant medial temporal lobe**
Visual/spatial memory poor; verbal learning relatively preserved	Visual reproduction/route memory	**Nondominant medial temporal network**

Educational bedside localisation aid. Use clinical context and appropriate investigations to confirm diagnosis.